Engineering Methodology

1. Multi-Agent Training Framework

- ★ **Phase 1: Training on Healthy Body**
  - Deep policy successfully performs **randomized tasks** given goals (rewards)
  - Learns **muscle control via neural drive signals**
- ★ **Phase 2: Policy Transfer to PD body**
  - PD modeling→ **error in tasks & movement**
- ★ **Phase 3: Multi-agent system:**
  - PD patient + exo agent
  - **Custom reward function** guides learning
    - "Good job, you restored x% motion!"

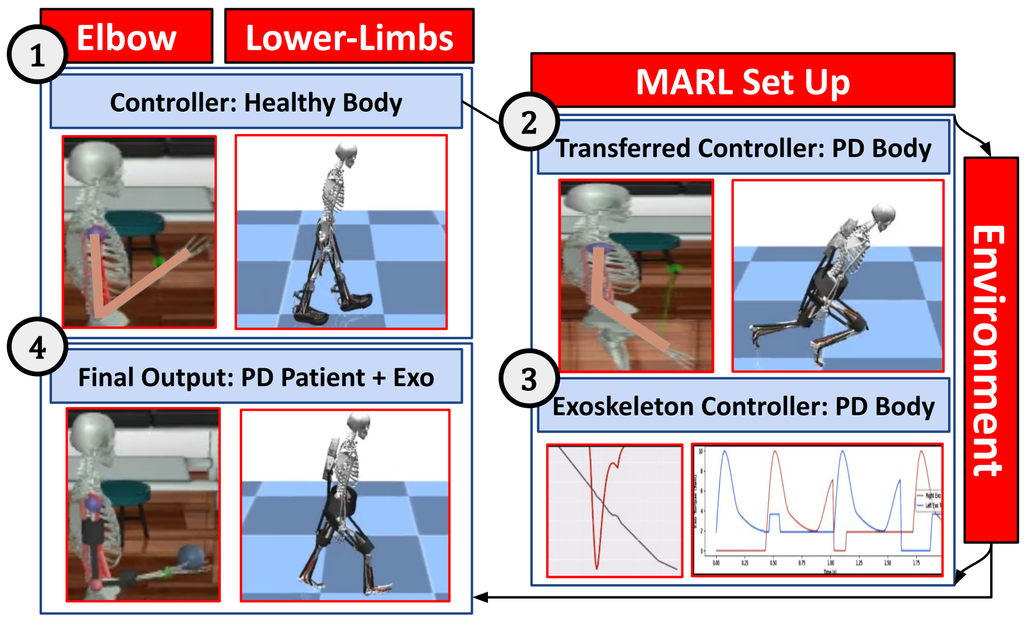


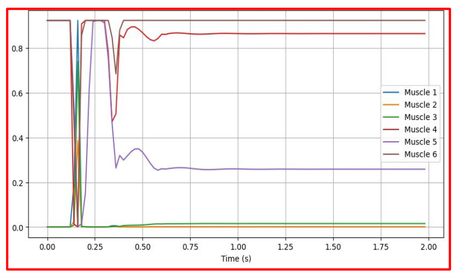
Figure 13: MARL Training Pipeline Flowchart utilizing Myosuite (18) & Myoassist (19) Simulators - Created by Researcher

2. Sensor Data

EMG

- ★ **Neural drive signals**
  - Derived from EMG:
  - Elbow → **6-m**, Legs → **22-m**
  - Measures **neural drive**$_{22}$
  - **Unnatural** bursts reflect bradykinesia
  - **Partly represents "intention"**

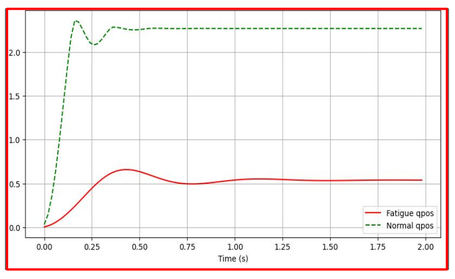
Figure 14: Muscle Excitation: Weak vs. Healthy - Created by Researcher



IMU

- ★ **Motion data** - joint kinematics
  - Derived from IMU sensors
    - Joint angle and velocity
  - Muscle degeneration + bradykinesia → **slower & weaker** IMU data
  - **Represents actual motion**

Figure 15: IMU: PD vs. Healthy - Created by Researcher



3. Training Environment Construction

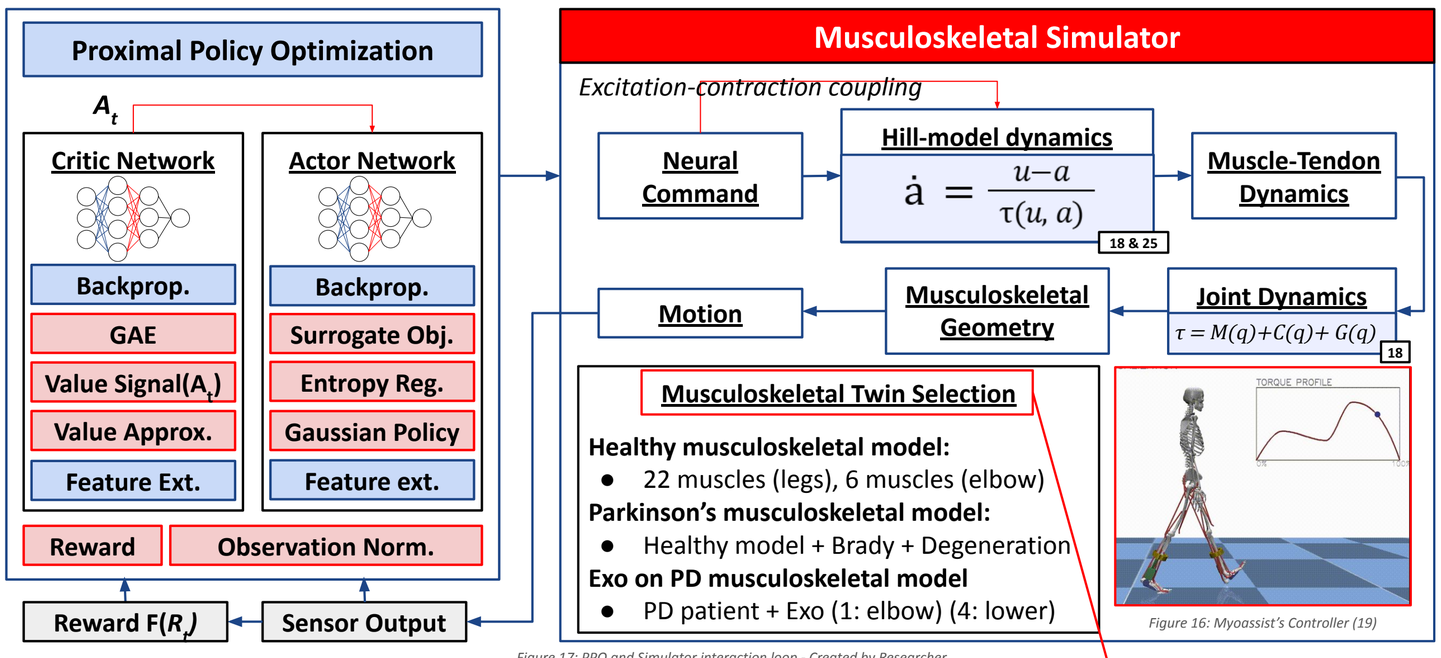
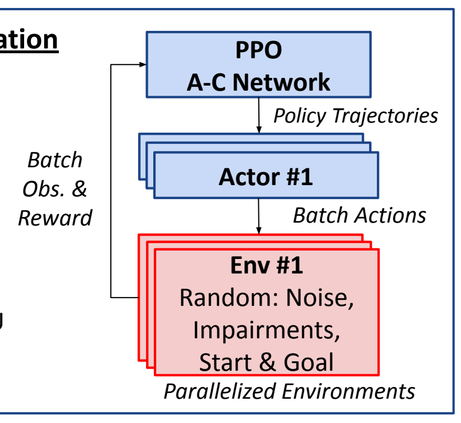


Figure 17: PPO and Simulator interaction loop - Created by Researcher

Figure 16: Myoassist's Controller (19)

Environment Vectorization

- ★ Multiple **random** simulators **run in parallel**
  - Efficient hyperparam tuning
  - Maximizes CPU efficiency



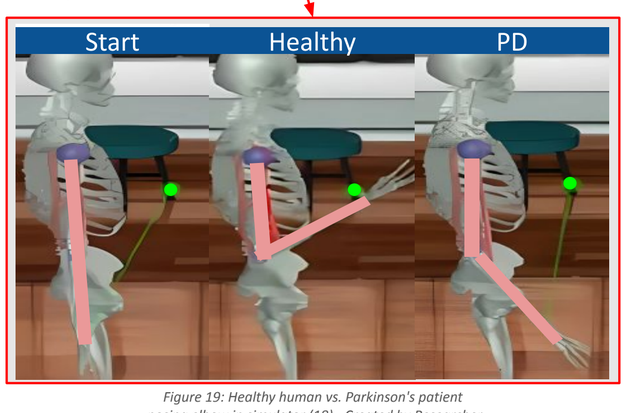


Figure 19: Healthy human vs. Parkinson's patient posing elbow in simulator (18) - Created by Researcher

Reward Function Design

$$R_t = -w_1(\Delta q_j)^2 - w_2 \Delta v_j \cdot (\Delta v_j > t) - w_3(\tau_j^{exo})^2 + w_4(\epsilon_t - \epsilon_{t-1})$$

1. **Target position penalty**
   - Reward for lower displacement from goal
   - Calculated relative to previous timestep
2. **Smoothness reward**
   - Penalizes sudden speed changes
3. **Energy efficiency reward**
4. **Restoration reward**
   - Parallel simulation: healthy model on same task compares alongside PD, exo-assisted motion

Training & Tuning

- ★ **Hyperparameter Tuning**
  - **Bayesian Optimization**
    - Probabilistic model selects hyperparams
  - **Median pruner**

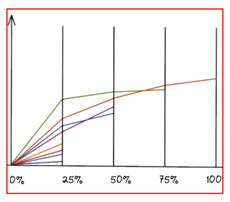


Figure 20: RL Pruner (24)

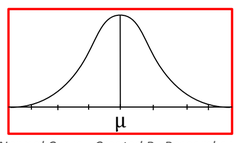
- ★ **Curriculum Learning**
  - **Reward weight tuning** → agent gradually builds upon previous experiences
- ★ Overcomes slow convergence

4. Model Configuration

- ★ **Adaptive Variability**
  - 400,000 episodes for training → **400,000 different patient instances**
  - **Domain randomization** → Each episode **randomizes** tasks and severity
- ★ **Critic**
  - **Value Estimation** → long-term reward
  - **Advantage Calculation (GAE)** → how much better the agent's action is than expected

Value Estimation
$V(s_t) = Q(r_t, s_t)$

- ★ **Stable Learning**
  - **Gaussian Policy** → outputs probability distribution
  - **Clipped Surrogate Objective** → stable policy updates
  - **Entropy Regularization** → exploration vs. exploitation



Normal Curve - Created By Researcher

- ★ **Feature Extractor Architecture**
  - Observation Space: Elbow - **8D** & Lower-limbs - **10-14D**
  - Temporal 2D CNN
    - Dual embedding path → filters capture **global & local** dynamics
    - **Attention to dominant activations** → eliminates noise → **richer obs.**
    - EMG-IMU Fusion → **context-aware control**

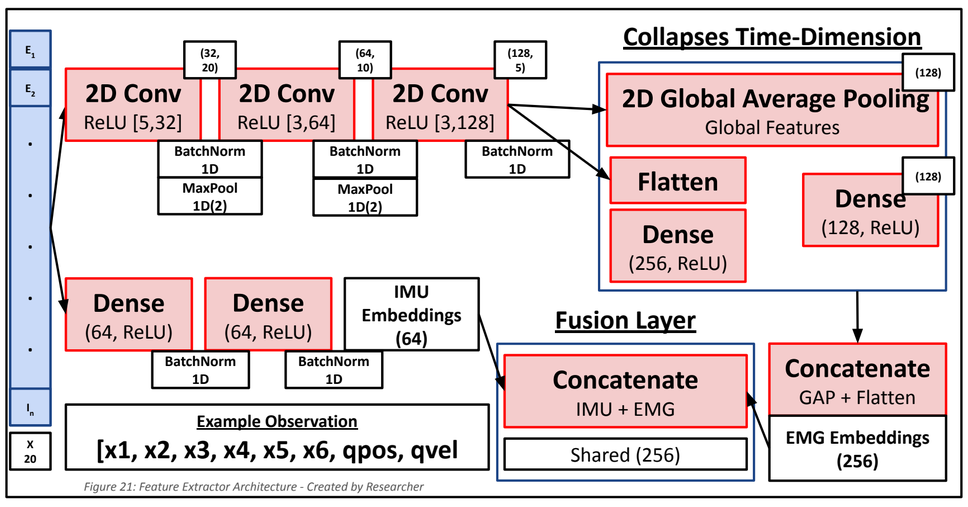


Figure 21: Feature Extractor Architecture - Created by Researcher

RLHF - Reinforcement Learning through Human Feedback

- ★ **User feedback interface**
- ★ **Personalization via metadata**
  - System tracks context (time, date, etc.)
  - Shapes device behavior
- ★ **Human-Aligned Learning**
  - Exoskeleton adapts to user's goals.
- ★ **Innovative Application**
  - Unique use of RLHF
  - Producer-user gap

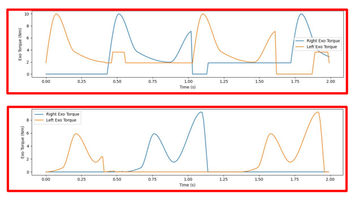


Figure 22-23: N vs. Smooth - Created by Researcher

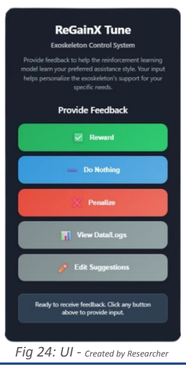


Fig 24: UI - Created by Researcher

Results

Symptom Monitoring

- ★ Exo assistance **correlates** with motor impairment severity
  - **Assistance varies patient-to-patient** → dynamic policy
- ★ Data is automatically logged by exo on ReGainX Tune app
- ★ Current methods: **subjective online tests**$_5$

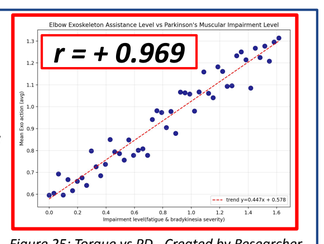


Figure 25: Torque vs PD - Created by Researcher

Cloud Exoskeleton System

- ★ **Sim-to-real transfer**
  - RegainX's models run on low-cost hardware, proving **viability**
  - **High-fidelity simulation** eliminates severe model **overfitting** to unrealistic dynamics.
    - Real-time results validate use of deep models in real envs
- ★ **Cloud-based operation** (reference fig 26)
  - Exo outputs torque every **0.1s** (5Hz)
    - Data is logged to **Firebase**
  - **Average cloud latency: ~0.029s** ~35Hz
- ★ **Low-Cost Design** - $130
  - **Arduino**(C++), **DC driver, and motor**
  - 12V battery → **daily assistance**
  - Easy-to-use straps + lightweight

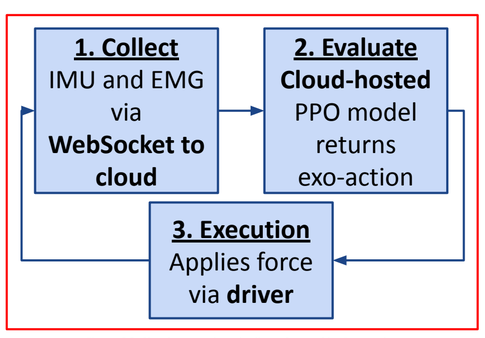


Figure 26: Cloud computing pipeline - Created by Researcher

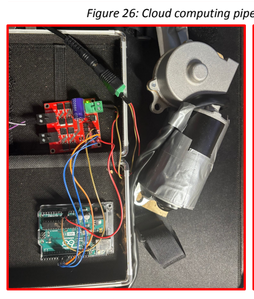


Figure 27: Hardware Pipeline - CBR

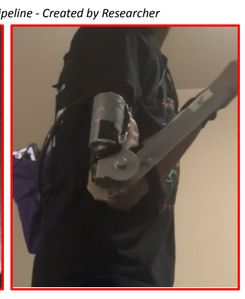


Figure 27: Researcher testing Elb-Exo

Accuracy Matrices

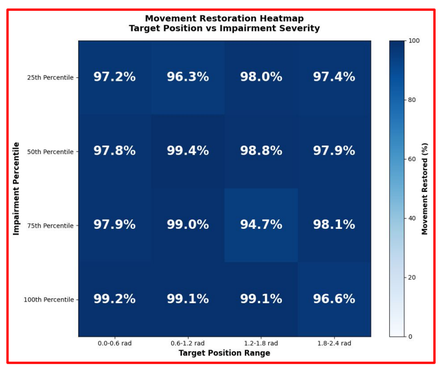


Figure 26: Elbow Exo Heatmap - Created by researcher

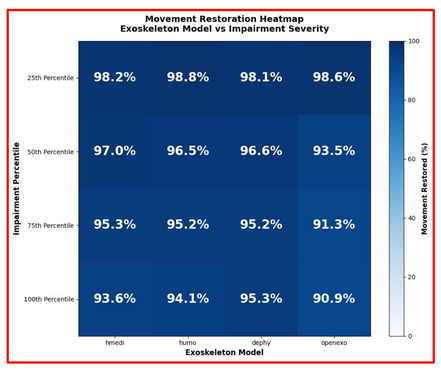


Figure 27: L-L Exo Heatmap - Created by researcher

- ★ **Elbow Exoskeleton**
  - Accuracy = **97.86%**
  - STDEV **= 2.72%**
- ★ **Lower-limb Exoskeletons**
  - Accuracy = **95.51%**
  - STDEV **= 2.34%**

Latency

- ★ **1.19ms** on ReGainX PPO
- ★ **Lower than frequency**

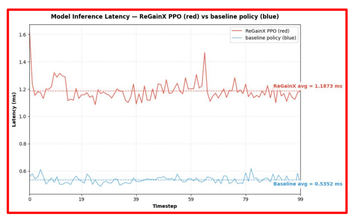


Figure 28: PPO vs baseline latency- Created by Researcher

Generalizability

- ★ **Stratified**, random evaluation methodology: 64,000 each.

Performance Graphs

Goal Convergence

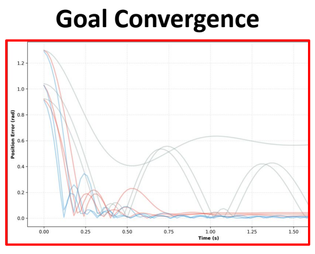


Figure 29: Created by Researcher

Average Muscle Action

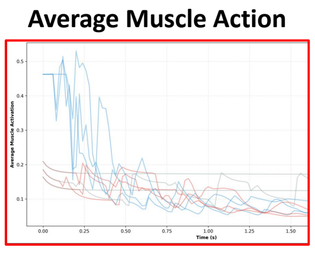


Figure 30: Created by Researcher

Restoration %

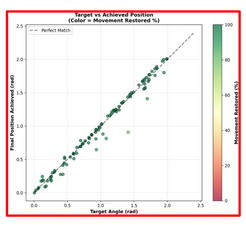


Figure 31: Created by Researcher

PD vs. EXO vs. Healthy

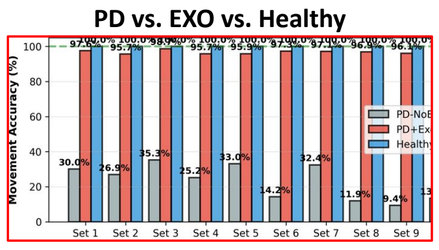


Figure 32: Created by Researcher